

Bill Psomas

Résumé

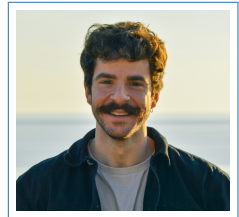
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Somewhere, something incredible is waiting to be known.

Summary

Dr. Bill Psomas is an early-career researcher in computer vision and machine learning. Most recently, he successfully defended his Ph.D. dissertation titled "Learning Visual and Multimodal Representations" at the National Technical University of Athens (NTUA). Prior to that, he received a Master of Science in "Data Science and Information Technologies" with a specialization in "Big Data and Artificial Intelligence" from the National Kapodistrian University of Athens (UoA) and a Diploma as a Rural, Surveying and Geoinformatics Engineer from NTUA. He has completed internships at research institutes such as Inria Rennes - Bretagne Atlantique and the Institute of Advanced Research in Artificial Intelligence. Dr. Psomas has research experience, having served as a research assistant at the Athena Research Center and NTUA. He has contributed to 2 European, 2 Greek, and 1 international research projects, co-supervised 5 Diploma/Master theses, and published 3 theses, 2 journal articles, and 6 conference and workshop papers in his fields of expertise. Additionally, he has organized 2 workshops and symposiums, served as a session chair at 1 conference, and reviewed papers for 3 journals and 5 conferences. Dr. Psomas has also contributed to the teaching of 3 undergraduate and 1 postgraduate courses at NTUA and has delivered 2 invited talks and lectures. Since 2024, he has been independently teaching the "Computer Vision" course for the Master's program in "Data Science and Information Technologies" at the University of Athens.

Research Interests

Computer vision, machine learning, Image classification, image retrieval, deep metric learning, clustering, nearest neighbor search, visual and multimodal representation learning, unsupervised and self-supervised learning, object detection, localization and recognition, explainability, deep network architectures, remote sensing, generative artificial intelligence.

Education

Degrees

- 2020–2024 **Ph.D. in Deep Learning for Computer Vision**, *National Technical University of Athens*, Greece
- 2018–2020 **MSc in "Data Science and Information Technologies"**, *National Kapodistrian University of Athens*, Greece, 9.6/10
Specialization: Big Data and Artificial Intelligence
- 2012–2018 **Diploma in Rural, Surveying and Geoinformatics Engineering**, *National Technical University of Athens*, Greece, 7.38/10
Specialization: Machine Learning for Remote Sensing
- 2009–2012 **High School Certificate**, *Nea Genia Ziridis*, Spata, Greece, 19.6/20
Specialization: Mathematics, Physics and Informatics

Summer Schools

- 2022 1st **ELLIS Doctoral Symposium**, *ELLIS, University of Tübingen, Max Planck Institute*, Tuebingen, Germany
- 2022 **Vision and Sports Summer School**, *Czech Technical University in Prague*, Czech Republic
- 2021 **PRAIRIE/MIAI Artificial Intelligence Summer School**, *PRAIRIE, MIAI, Naver Labs, Inria*, Virtual

Other

- 2017– **Online Courses**, *Coursera, edX, Udacity*
I have attended and completed over 12 online courses covering a broad range of topics, including Artificial Intelligence, Machine/Deep Learning, Neural Networks, Image Processing, Computer Vision, Natural Language Processing and more.

Experience

Research

- 2024– **Postdoctoral Researcher**, *Czech Technical University in Prague*, Czech Republic
- **Research Directions:**
 - Instance-level generation.
 - Open-vocabulary segmentation.
 - Composed image retrieval.
- 2024– **Visiting Lecturer**, *National Kapodistrian University of Athens*, Greece
- **Course:** Teaching "Computer Vision" for Master of Science in "Data Science and Information Technologies".
- 2024–2024 **Postdoctoral Researcher**, *National Technical University of Athens*, Greece
- **Project:** "RAMONES: Radioactivity Monitoring in Ocean Ecosystems", FET-H2020.
 - Contributed to WP1: "Design and develop novel underwater gamma radiation instruments".
 - Contributed to WP4: "Multi-modal data analysis and environmental modeling".

- 2020–2024 **Doctoral Researcher**, *National Technical University of Athens*, Greece
- **Project:** “RAMONES: Radioactivity Monitoring in Ocean Ecosystems”, FET-H2020.
 - Contributed to WP1: “Design and develop novel underwater gamma radiation instruments”.
 - Contributed to WP4: “Multi-modal data analysis and environmental modeling”.
 - **Project:** “BiCUBES: Analysis-Ready Geospatial Big Data Cubes and Cloud-based Analytics for Monitoring Efficiently our Land & Water”, HFRI.
 - Contributed to WP1: “Data Harmonization and Cloud-Free Mosaics”.
 - Contributed to WP4: “Change Detection and Time Series Analytics through Recurrent Networks”.
 - **Project:** “iTOBOS: Intelligent Total Body Scanner for Early Detection of Melanoma”, H2020.
 - Contributed to WP7: “Integration of Explainable AI for Data-Feature and Model Quality Control”.
 - Contributed to WP8: “AI-based Multimodal Data Integration”.
 - Contributed to WP9: “Integration and Technical Validation”.
 - Contributed to WP12: “Communication, Dissemination”.
 - **Project:** “eOSD: An Intelligent Early-Warning Oil Spill Detection and Prediction System for the Arabian Gulf and the Red Sea”, KAUST.
 - Contributed to T1 and T3: “Data collection, algorithm development for detection, and validation of the results”.
- 2022–2023 **Research Intern**, *Institute of Advanced Research in Artificial Intelligence*, Vienna, Austria
- Developed an attention-based pooling mechanism enhancing both convolutional networks and vision transformers.
- 2020–2020 **Research Intern**, *Inria Rennes - Bretagne Atlantique*, Rennes, France
- Developed a novel evaluation setting and a novel loss function for deep metric learning.
- 2019–2020 **Research Assistant**, *Athena Research Center*, Marousi, Attiki, Greece
- **Project:** “AdVENT: Augmented Visitor Experience in National Parks”, NSRF 2014-2020.
 - Contributed to WP4: “Development of an application for plant species recognition in forests using machine learning and computer vision techniques”.

Other

- 2020– **Computer Vision Engineer**, *Mismatch*, Patras, Greece
- Designed background subtraction algorithm for basketball court images.
 - Trained efficient object detectors for player/ball detection.
 - Developed automatic pan and zoom algorithm for camera movement.
 - Trained efficient object detectors for digit detection on basketball boards.
 - Created jersey color and jersey number datasets using vision-language models.
 - Trained models for spatio-temporal action localization.
- 2018–2019 **Software Developer**, *MaP Ltd.*, Holargos, Attiki, Greece
- Developed a coordinate recognition and drawing system.
 - Implemented CNNs for face and license plate detection and censoring.
 - Developed the CrossClimate web map application.

Miscellaneous

2015–2018 **Freelance Photographer**, Athens, Greece

- Photographed products for commercial use, including fashion items and other merchandise.
- Captured interior shots for short-term rentals and real estate listings.
- Provided professional photography services for weddings and other ceremonies.
- Conducted photo editing and retouching to ensure high-quality final images.

Professional memberships

2024- Institute of Electrical and Electronics Engineers [IEEE].

2018- Technical Chamber of Greece [TCG].

Teaching

Independent

2024- **Computer Vision**, *National Kapodistrian University of Athens*, Greece

Non-independent

2021-2024 **Advanced Remote Sensing**, *National Technical University of Athens*, Greece

2023-2024 **Remote Sensing II**, *National Technical University of Athens*, Greece

2023-2024 **Remote Sensing I**, *National Technical University of Athens*, Greece

2022-2023 **Special Topics of Remote Sensing**, *National Technical University of Athens*, Greece

Supervision

Non-independent

2024 **Despoina Tsigarida**, *National Technical University of Athens*, Exploring Pooling Methods in Deep Metric Learning: A Performance Analysis on Benchmark Datasets

2024 **Eirini Baltzi**, *National Technical University of Athens*, Attentive Probing: Leveraging Attention for Improved Feature Utilization in Self-Supervised Models

2024 **Kalliopi Karadima**, *National Technical University of Athens*, Crater Detection on Mars using Deep Learning Techniques

2022 **Yannis Tsiotas**, *National Technical University of Athens*, Utilizing Self Supervised Methods in Unsupervised Metric Learning

2022 **Ioanna Gogousou**, *National Technical University of Athens*, Studying and Evaluating Self-Supervised Learning Methods for Image Classification

Scientific Activities

Invited Lectures/Talks

2024 **Machine Learning for Earth Observation**, *International Geoscience and Remote Sensing Symposium Summer School [IGARSSS]*, 3 hour lecture and lab

2024 **Leveraging Attention in Masked Image Modeling and Pooling**, 48th *Pattern Recognition and Computer Vision Colloquium [PRCVC]*, 30 minutes talk

[Chair](#)

2024 **Session TH2.R2: Classification and Clustering II**, *International Geoscience and Remote Sensing Symposium [IGARSS]*, Co-chair

[Organizer](#)

2023 13th **Workshop on Hyperspectral Imaging and Signal Processing: Evolution in Remote Sensing [whispers]**, *IEEE GRSS, DLR DESIS*, Student team

2023 2nd **ELLIS Doctoral Symposium [EDS]**, *ELLIS Alicante*, Student team

[Reviewer in journals](#)

2023- **International Journal of Computer Vision [IJCV]**

2022- **Transactions on Neural Networks and Learning Systems [TNNLS]**

2022- **Transactions on Pattern Analysis and Machine Intelligence [TPAMI]**

[Reviewer in conferences](#)

2024- **European Conference on Computer Vision [ECCV]**

2024- **International Geoscience and Remote Sensing Symposium [IGARSS]**

2023- **Conference on Computer Vision and Pattern Recognition [CVPR]**

2023- **Winter Applications on Computer Vision [WACV]**

2022- **Advances in Neural Information Processing Systems [NeurIPS]**

[Honors & Awards](#)

2019 **Finalist in Industry 4.0 Crowdhackathon**, *Hellenic Federation of Enterprises, Crowdpolicy*, Smart Glasses Application

2019 **Finalist in ESA Space App Camp**, *ESA ESRIN*, AI generated multi-player mini-games

2018 3rd **place in Copernicus Hackathon in Athens**, *ESA, Corallia*, VR Video Game using Copernicus data

[Grants](#)

[Non-independent](#)

2023 **Computational time grant**, *Greek HPC facility ARIS*, project PR015025: National Scale Multi-annual Land Cover and Crop type Mapping for Greece

2021 **Computational time grant**, *Greek HPC facility ARIS*, project PR009017: Continual Learning for Retrieving and Recognizing Landmarks

Publications

Conference Papers

- [1] Ioannis Kakogeorgiou, Spyros Gidaris, Bill Psomas, Yannis Avrithis, Andrei Bursuc, Konstantinos Karantzas, and Nikos Komodakis. What to hide from your students: Attention-guided masked image modeling. In *European Conference on Computer Vision*, 2022.
- [2] Bill Psomas, Ioannis Kakogeorgiou, Nikos Efthymiadis, Giorgos Tolias, Ondrej Chum, Yannis Avrithis, and Konstantinos Karantzas. Composed image retrieval for remote sensing. In *IEEE International Geoscience and Remote Sensing Symposium*, 2024.
- [3] Bill Psomas, Ioannis Kakogeorgiou, Konstantinos Karantzas, and Yannis Avrithis. Keep it simple: Who said supervised transformers suffer from attention deficit? In *International Conference on Computer Vision*, 2023.
- [4] Piera Riccio, Bill Psomas, Francesco Galati, Francisco Escolano, Thomas Hofmann, and Nuria M Oliver. Openfilter: A framework to democratize research access to social media AR filters. In *Advances in Neural Information Processing Systems*, 2022.
- [5] Simon Vellas, Bill Psomas, Kalliopi Karadima, Dimitrios Danopoulos, Alexandros Paterakis, George Lentaris, Dimitrios Soudris, and Konstantinos Karantzas. Evaluation of resource-efficient crater detectors on embedded systems. In *IEEE International Geoscience and Remote Sensing Symposium*, 2024.
- [6] Shashanka Venkataramanan, Bill Psomas, Ewa Kijak, Laurent Amsaleg, Konstantinos Karantzas, and Yannis Avrithis. It takes two to tango: Mixup for deep metric learning. In *International Conference on Learning Representations*, 2022.

Journal Articles

- [7] Elisa Arnone, George Stavridis, Livia Ortolani, Athanasios Rammos, Marco Cucchi, Anthoula Gkiourou, Sara Dal Gesso, Marco Petitta, and Bill Psomas. The drought-alert decision support system for water resources management. *Desalination and Water Treatment*, 194:304–314, 2020.
- [8] Maria Sdraka, Ioannis Papoutsis, Bill Psomas, Konstantinos Vlachos, Konstantinos Ioannidis, Konstantinos Karantzas, Ilias Gialampoukidis, and Stefanos Vrochidis. Deep learning for downscaling remote sensing images: Fusion and super-resolution. *IEEE Geoscience and Remote Sensing Magazine*, 2022.

Theses

- [9] Bill Psomas. Investigation of greek character automatic recognition methods in ancient epigraphic images, February 2018.
- [10] Bill Psomas. Metric learning: A deep dive, July 2020.
- [11] Bill Psomas. Learning visual and multimodal representations, September 2024.